

Folland Real Analysis 2nd

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1.2.3. mid-term. 5.6.8 Section.

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Notation. $\mathbb{N} = \mathbb{Z}_{>0}$ $\overline{\lim} E_n = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_n$ i.e. $x \in E_n$ for infinitely many n
 $\underline{\lim} E_n = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_n$ i.e. $x \in E_n$ for all but finitely many n .

Partial ordering: ① $x \geq y, y \geq z \Rightarrow x \geq z$
 ② $x \geq y$ and $y \geq x \Rightarrow x = y$
 ③ $x \geq x$.

If $\forall x, y \in X, x \geq y$ or $y \geq x$, it is called total ordering.

maximal element: $x: \forall y \in X, x \leq y \Rightarrow x = y$

Hausdorff Maximal Principle: Every partially ordered set has a maximal total ordered set.

Zorn's Lemma: If X is a partially ordered set and every totally ordered subset of X has an upper bound, then X has a maximal element.

The axiom of choice: $\{X_\alpha: \alpha \in A\}$ is a non-empty collection of non-empty sets, then $\prod_{\alpha \in A} X_\alpha$ is non-empty.

Corollary. If $\{X_\alpha: \alpha \in A\}$ is a disjoint collection of nonempty sets, there's a set $Y \subset \bigcup_{\alpha \in A} X_\alpha$ s.t. $Y \cap X_\alpha$ contains precisely one element for each $\alpha \in A$.

$$(X_n). \quad \overline{\lim} X_n = \inf_{k \geq 1} (\sup_{n \geq k} X_n)$$

$$\underline{\lim}_{n \rightarrow \infty} X_n = \sup_{k \geq 1} (\inf_{n \geq k} X_n)$$

uncountable sum of nonnegative number.
 $f: X \rightarrow [0, +\infty)$.

$$\sum_{x \in X} f(x) := \sup \left\{ \sum_{x \in F} f(x) : F \subset X, F \text{ finite} \right\}.$$

Let $A = \{x \in X : f(x) > 0\}$.

If A uncountable: $\sum_{x \in X} f(x) = +\infty$.

If A countable: 可任意排序为级数求和

Proposition. Every open set in $\mathbb{R}$ is a countable disjoint union of open intervals.

Thm. If E is a subset of the metric space (X, f) , TFAE.

① E is complete and totally bounded $\forall \varepsilon > 0, \exists$ finitely many balls of radius ε cover E .
 ② (Bolzano-Weierstrass Property) Every sequence in E has a subsequence that converges to a point of E .

③ (Heine-Borel Property) If $\{V_\alpha: \alpha \in A\}$ is a cover of E by open sets, $\exists$ a finite set $F \subset A$ s.t. $\{V_\alpha: \alpha \in F\}$ covers E .

n -dim subset "volume". (直感に訴へた目指し)

$$\mu: \mathcal{P}(E) \rightarrow [0, +\infty]$$

i. if $E_1, E_2, \dots$ is a finite or infinite sequence of disjoint set, then

$$\mu(E_1 \cup E_2 \cup \dots) = \mu(E_1) + \mu(E_2) + \dots$$

ii. If E is congruent to F , then $\mu(E) = \mu(F)$

iii. $\mu(Q) = 1$, where Q is the unit cube.

$$[0,1) \quad x \sim y \Leftrightarrow x-y \in \mathbb{Q}.$$

Let N be the 代表元素. clearly infinite.

$$\mu([0,1]) = \mu(N) + \mu([0,1] \setminus N) \Rightarrow \mu(N) \leq \mu([0,1]) = 1.$$

$$\mu([0,$$